Amit Karthikeyan

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Education

University of California, Santa Barbara — Santa Barbara, CA
Bachelor of Science in Computer Science, Minor in Statistical Science — Expected Grad Date: June 2027
GPA: 3.73 — Dean's List
Relevant Coursework: Data Structures & Algorithms, Object Oriented Design, Probability & Statistics

Skills

Languages: Python, SQL, Java, C, C#, C++, TypeScript, JavaScript, Golang, HTML/CSS
Frameworks and Libraries: React, Node.js, Express.js, Next.js, FastAPI, PyTorch, NumPy, Pandas
Tools: Git, GitHub, AWS, Google Cloud, Linux/Unix, Jira, Docker, Excel, CI/CD
Databases: PostgreSQL, Pinecone, Firebase Firestore, DynamoDB, Redis

Experience

AutoInvent — Jul 2025 – Present
Founding Software Engineer — Remote

- Built an MVP for an AI powered legal drafting that helped secure $500K in pre-seed funding for development
- Architected a RAG pipeline using Google Gemini, Hugging Face Transformers, and Pinecone to automate patent claim and specs generation, processing unstructured text inputs with <2 seconds latency per request
- Developed a scalable React 18 + Vite frontend with real-time document updates and responsive UI
- Engineered a multi-tenant data layer using Firestore + Cloud KMS, enforcing tenant-scoped access controls for strict data isolation and legal compliance

DMS Solutions — Aug 2023 – June 2024
Software Engineer Intern — Simi Valley, CA

- Developed an e-commerce website using JavaScript and HTML resulting in $20k+ in online sales
- Improved UI and site navigation to increase product discoverability, contributing to 1,000+ products sold
- Integrated payments, shipping, tax logic, and basic analytics to track conversions and sales performance

Projects

Mapache | Golang, Python, Docker, React, TypeScript

- Designed a real-time ingestion service in Go handling 2,000+ events/s with <250ms end-to-end latency
- Implemented a ClickHouse schema optimized for time-series data, no observed data loss across 10M+ messages
- Built a NumPy-vectorized CAN-bus decoder in Python, reducing parsing CPU overhead by 30%

SignalBench | Python, FastAPI, PostgreSQL, Redis, React

- Benchmarked logistic regression, gradient boosting, and neural baselines with ROC-AUC and calibration analysis
- Stored 70+ experiment runs, configurations, and evaluation metrics in a relational SQL database with PostgreSQL
- Implemented Redis-backed caching and rate limiting to reduce external data fetches and protect training endpoints
- Built a FastAPI backend to host RESTful APIs and handle query logic for model training, predictions, and metrics

Factify | GCP, React, JavaScript, Next.js, HTML/CSS

- Shipped a Chrome extension in JavaScript that fact-checks social media content, processing 250+ posts daily
- Optimized API performance by 85% through intelligent request batching, using combined analysis techniques
- Cut LLM token usage by 60–80% with prompt engineering, strict length caps, and structured JSON output
- Integrated Stripe subscriptions and webhooks to enforce tiered usage limits and automate billing events

Leadership

Gaucho Racing - Formula SAE — Sep 2024 - Present
Data Science Lead — Santa Barbara, CA

- Led end-to-end telemetry analytics for the UCSB Formula SAE team with over 20 members
- Built Python/SQL pipelines and ClickHouse dashboards to model lap time, driver performance, and reliability
- Created collaboratively in a SCRUM environment and delivered data driven race strategy recommendations